8^{th} Grade Unit 4 Assessment Guide

General Information	Students have been building their understanding of the Laws of Exponents since grade 6. Unit 1 reviewed the Laws of Exponents students learned in grades 6 and 7. As students often struggle with negative exponents, it is important that prior to Unit 4 students are fluent with the Laws of Exponents they learned in grades 6 and 7. In Unit 3, students used negative exponents when writing numbers using scientific notation and performing operations with numbers written in scientific notation. The scope and sequence was specifically designed so that students had multiple learning opportunities to grasp negative exponents. The draft test item specifications state that items for MA.8.NSO.1.3 must use negative exponents. Since this is the only new Law of Exponents learned in grade 8 this assessment limit allows items for statewide assessment to discern grade 8 knowledge from grade 6 and grade 7. All of the Laws of Exponents learned in grades 6 and 7 are considered prior knowledge and will not be considered on-grade-level learning. In Algebra 1, students will extend the Laws of Exponents to include rational exponents. The assessment guide is designed to help teachers discern which Laws of Exponents students may be struggling on. The guide includes the names of the laws of exponents that each question assesses. Note that students may take different approaches when generating equivalent expressions or evaluating exponential expressions, so these notes on the exponents used are not meant to indicate there is only one way. Students may utilize different laws depending on their approach. Teachers may want to use this feature to note which laws students are struggling on so that support can be more focused. Teachers may wish to use this information to better inform any remediation students may benefit from if students are showing that they struggle with the Laws of Exponents learned in grade 6 (MA.6.NSO.3.3) and grade 7 (MA.7.NSO.1.1).
Calculator Information	The questions on the unit assessment are no calculator.
Total Recommended Points	The total recommended points in this guide are 100 points

Question	1	Benchmark(s)	MA.8.NSO.1.3
☐ $\frac{1}{64}$ ☐ $(-8)^{-8}$ ☐ $(-8)^{-2}$ ☐ $\frac{(-8)(-8)(-8)(-8)}{(-8)(-8)}$ ☐ $\frac{(-8)(-8)}{(-8)(-8)(-8)(-8)}$ ☐ $\frac{1}{(-8)(-8)(-8)(-8)(-8)(-8)(-8)(-8)}$		**Recommended Scoring** **(12 Total Points)**	Consider giving students 4 points for each equivalent expression.
		Law(s) of Exponents	• Product of Powers • Negative Exponent

Question	2	Benchmark(s)	MA.8.NSO.1.3
☐ $-1\left(\frac{1}{2\cdot 3}\right)^{1}$ ☐ $-\frac{1}{6}$ ☐ $\left(\frac{6}{1}\right)^{1}$ ☐ $\frac{1^{-1}}{6^{-1}}$ ☐ 6		**Recommended Scoring** **(12 Total Points)**	Consider giving students 4 points for each equivalent expression.
		Law(s) of Exponents	• Negative Exponent • Power of a Quotient • Identity Exponent

Question	3	Benchmark(s)	MA.8.NSO.1.3
b^x where x is a positive number: $(-5)^4$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Product of Powers

Question	4	Benchmark(s)	MA.8.NSO.1.3
b^x where x is a positive number: $\left(\frac{7}{12}\right)^3$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Negative Exponent

Question	5	Benchmark(s)	MA.8.NSO.1.3
b^x where x is a positive number: $\left(\frac{11}{6}\right)^{24}$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Power of a Power • Negative Exponent

Question	6	Benchmark(s)	MA.8.NSO.1.3
b^x where x is a positive number: $\left(\frac{1}{13}\right)^6$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Quotient of Powers • Negative Exponent

Question	7	Benchmark(s)	MA.8.NSO.1.3
		Recommended Scoring (4 Total Points)	No partial credit should be given.
b^x where x is a positive number: $-\frac{1}{18}$		Law(s) of Exponents	• Negative Exponent • Identity Exponent
Question	**8**	**Benchmark(s)**	MA.8.NSO.1.3
		Recommended Scoring (4 Total Points)	No partial credit should be given.
b^x where x is a positive number: $\left(\frac{1}{37}\right)^{11}$		Law(s) of Exponents	• Product of Powers • Negative Exponent • Identity Exponent
Question	**9**	**Benchmark(s)**	MA.8.AR.1.1
A. $-1 \cdot y$ B. $-\frac{1}{y}$ C. y D. $\frac{1}{y}$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Negative Exponent
Question	**10**	**Benchmark(s)**	MA.8.NSO.1.3
		Recommended Scoring (4 Total Points)	No partial credit should be given.
value: 1.331		Law(s) of Exponents	• Product of Powers • Quotient of Powers

Question	11	Benchmark(s)	MA.8.NSO.1.3
value: $-\frac{1}{120}$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Power of a Quotient • Negative Exponent • Quotient of Powers
Question	12	Benchmark(s)	MA.8.NSO.1.3
value: $\frac{8}{125}$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Zero Exponent • Negative Exponent • Power of a Quotient
Question	13	Benchmark(s)	MA.8.NSO.1.3
value: $\frac{169}{81}$		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Power of a Power • Product of Powers • Power of a Quotient • Negative Exponent
Question	14	Benchmark(s)	MA.8.AR.1.1
equivalent expression: m^{22}		Recommended Scoring (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Product of Powers • Power of a Power

Question	15	Benchmark(s)	MA.8.AR.1.1
equivalent expression	$\frac{1}{x^2}$	**Recommended Scoring (4 Total Points)**	No partial credit should be given.
		Law(s) of Exponents	• Power of a Power • Quotient of Powers • Negative Exponent
Question	**16**	**Benchmark(s)**	MA.8.AR.1.1
equivalent expression	$\frac{1}{y^2}$	**Recommended Scoring (4 Total Points)**	No partial credit should be given.
		Law(s) of Exponents	• Product of Powers • Quotient of Powers • Negative Exponent
Question	**17**	**Benchmark(s)**	MA.8.AR.1.1
equivalent expression	$\frac{w^8}{16}$	**Recommended Scoring (4 Total Points)**	No partial credit should be given.
		Law(s) of Exponents	• Power of a Quotient • Quotient of Powers • Negative Exponent
Question	**18**	**Benchmark(s)**	MA.8.AR.1.1
equivalent expression	$\frac{28y^{11}}{x^4}$	**Recommended Scoring (4 Total Points)**	No partial credit should be given.
		Law(s) of Exponents	• Product of Powers • Negative Exponent

Question	19	Benchmark(s)	MA.8.AR.1.1
equivalent expression: $a^{28}b^{20}$		**Recommended Scoring** (4 Total Points)	No partial credit should be given.
		Law(s) of Exponents	• Quotient of Powers • Power of a Product

Question	20	Benchmark(s)	MA.8.AR.1.1
☐ x^{14} ☐ x^{20} ☐ $\left(\frac{x^{-13}}{x^{11}} \cdot x^{10}\right)$ ☐ $\left(\frac{x^{26}}{x^{-22}} \cdot x^{10}\right)$ ☐ $\left(\frac{1}{x^{24}} \cdot x^{-5}\right)^{-2}$ ☐ $(x^{-2} \cdot x^{-5})^{-2}$		**Recommended Scoring** (8 Total Points)	Consider giving students 4 points for each equivalent expression.
		Law(s) of Exponents	• Quotient of Powers • Power of a Product